

Doctor Profile & Notifications

Frontend Developer Guide

Version: 2.0 | Date: July 2026 | Audience: Frontend developers

Backend: HM-System (FastAPI) | Auth: JWT Bearer token

Modules: /doctor/profile + /doctor/notifications

Matches current nested API schemas (address, emergency_contact, department, role, shift).

1. Purpose & Scope

This document describes the Doctor Profile and Doctor Notifications APIs so frontend developers can implement the doctor UI correctly. Use this guide as the source of truth for endpoints, payloads, validation, permissions, and UI behaviour.

What this covers

- Doctor self-service profile: view, update, upload/delete profile image
- Which fields doctors can edit vs which are admin-only (read-only in UI)
- Nested payload shapes (address, emergency_contact, department, role, shift)
- Profile completion boolean + completion percentage
- In-app doctor notifications: list, unread badge, mark one/all as read
- Notification types, priorities, filters, and deep-link hints
- Suggested TypeScript types and frontend implementation checklist

What this does NOT cover

- Admin staff APIs (/users) except where they own doctor fields
- Push notifications, WebSocket, or SSE (not implemented - use polling)
- Delete notification or notification preferences (not implemented)

Auth header (all endpoints)

```
Authorization: Bearer <access_token>
```

Login via POST /auth/login. JWT must include the required permission. Inactive accounts receive 403. Soft-deleted / missing users receive 404.

2. Module A - Doctor Profile

Prefix: /doctor | Router: Routers/doctor_profile_router.py | Tag: Doctor Profile

Only users with role name "doctor" can use these endpoints. Admins manage license, fee, specialization, department, employee_id, joining_date, and shift via /users - not via /doctor/profile.

2.1 Endpoint summary

Method + Path	Status	Permission	Purpose
GET /doctor/profile	200	doctor_profile:view	Load full profile
PUT /doctor/profile	200	doctor_profile:update	Update editable fields
POST /doctor/profile/image	200	doctor_profile:upload_image	Upload avatar

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DELETE /doctor/profile/image	200	doctor_profile:delete_image	Remove avatar
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2.2 GET /doctor/profile

Request body: none. Returns DoctorProfileResponse (nested objects).

```
{
  "user_id": 12,
  "first_name": "Asha",
  "last_name": "Mehta",
  "email": "asha.mehta@hospital.com",
  "phone": "9876543210",
  "phone_code": "+91",
  "address": {
    "line": "12 MG Road",
    "city": "Pune",
    "state": "Maharashtra"
  },
  "date_of_birth": "1985-04-12",
  "gender": 2,
  "emergency_contact": {
    "name": "Raj Mehta",
    "phone": "9123456780"
  },
  "department": { "id": 3, "name": "Cardiology" },
  "role": { "id": 5, "name": "doctor" },
  "specialization": "Interventional Cardiology",
  "qualification": "MD, DM Cardiology",
  "medical_license_number": "MH-MED-12345",
  "employee_id": "DOC-0012",
  "experience_years": 12,
  "joining_date": "2018-06-01",
  "consultation_fee": 800.0,
  "bio": "Consultant cardiologist with 12 years experience.",
  "languages": ["English", "Hindi", "Marathi"],
  "shift": {
    "name": "Morning",
    "start_time": "09:00",
    "end_time": "13:00"
  },
  "profile_image_url": "/uploads/doctor_image/a1b2c3d4.jpg",
  "is_profile_completed": true,
  "profile_completion_percentage": 86,
  "is_active": true,
  "last_login": "2026-07-15T09:00:00+05:30",
  "created_at": "2026-01-10T10:00:00+05:30",
  "updated_at": "2026-07-14T18:20:00+05:30"
}
```

Response field reference:

Field	Type	Notes
user_id	number	Doctor user id
first_name	string	Read-only for doctor
last_name	string null	Read-only for doctor
email	string (email)	Read-only for doctor

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phone	string null	Editable
phone_code	string null	Editable (e.g. +91)
address	object	{ line, city, state } - editable
date_of_birth	string null	YYYY-MM-DD, editable
gender	number null	1-4, see Gender codes
emergency_contact	object	{ name, phone } - editable
department	object null	{ id, name } - read-only
role	object null	{ id, name } - read-only
specialization	string null	Admin-owned. Read-only
qualification	string null	Editable
medical_license_number	string null	Admin-owned. Read-only
employee_id	string null	Admin-owned. Read-only
experience_years	number null	Editable, 0-60
joining_date	string null	Admin-owned. Read-only
consultation_fee	number null	Admin-owned. Read-only
bio	string null	Editable
languages	string[]	Editable, default []
shift	object null	{ name, start_time, end_time } - read-only
profile_image_url	string null	Public path under /uploads
is_profile_completed	boolean	Computed server-side (strict)
profile_completion_percentage	number	0-100 progress bar value
is_active	boolean	Account status
last_login	datetime null	ISO-8601
created_at / updated_at	datetime null	Profile timestamps

Gender codes (integer, not label)

- 1 = Male
- 2 = Female
- 3 = Other
- 4 = Prefer not to say

API returns the integer. Frontend should map to labels in the UI.

Errors

- 401 - Missing/invalid JWT
- 403 - Missing permission, inactive account, or role is not doctor
- 404 - User/profile not found ("Doctor profile not found. Contact admin.")

2.3 PUT /doctor/profile

Content-Type: application/json. Schema: DoctorProfileUpdate with extra="forbid". Unknown/admin-only fields cause 422. At least one field is required; empty body -> 400.

All fields optional; send only what changed. Nested objects are partial (merge by key):

```
{
  "qualification": "MD, DM Cardiology",
  "experience_years": 12,
  "bio": "Consultant cardiologist...",
```

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```
{
  "languages": ["English", "Hindi", "Marathi"],
  "phone": "9876543210",
  "phone_code": "+91",
  "address": {
    "line": "12 MG Road",
    "city": "Pune",
    "state": "Maharashtra"
  },
  "date_of_birth": "1985-04-12",
  "gender": 2,
  "emergency_contact": {
    "name": "Raj Mehta",
    "phone": "9123456780"
  }
}
```

Validation rules:

- qualification: max 255 chars
 - experience_years: integer, >= 0 and <= 60
 - phone / emergency_contact.phone: max 20 chars
 - phone_code: max 10 chars
 - address.city / address.state: max 100 chars
 - emergency_contact.name: max 120 chars
 - gender: integer 1-4
 - languages: trimmed; empties dropped; case-insensitive dedupe
 - Do NOT send flat city/state/address strings - use nested address object
 - Do NOT send: first_name, last_name, email, department, role, specialization, medical_license_number, employee_id, joining_date, consultation_fee, shift, profile_image_url, is_profile_completed, profile_completion_percentage
- Response: same DoctorProfileResponse as GET. Side effect: is_profile_completed and profile_completion_percentage recomputed.

Profile completion rules

is_profile_completed is true only when ALL of these are set: qualification (truthy), experience_years (not null - 0 counts), bio (truthy). Image, languages, phone, license, and fee are NOT required for this boolean.

profile_completion_percentage (0-100) is a separate progress metric based on 14 checks: phone, phone_code, address line, city, state, DOB, gender, emergency name, emergency phone, qualification, experience_years, bio, languages (non-empty), profile image. Use this for a progress bar; use is_profile_completed for the incomplete banner / gate.

UI field ownership (important)

- EDITABLE by doctor: qualification, experience_years, bio, languages, phone, phone_code, address.{line,city,state}, date_of_birth, gender, emergency_contact.{name,phone}, profile image
- READ-ONLY display: first_name, last_name, email, department, role, specialization, medical_license_number, employee_id, joining_date, consultation_fee, shift, is_active, is_profile_completed, profile_completion_percentage
- Show a short note that license / fee / specialization / department / shift are managed by admin

2.4 POST /doctor/profile/image

Content-Type: multipart/form-data. Field name MUST be: file

- Allowed extensions: .jpg, .jpeg, .png, .webp
- Max size: 5 MB
- Empty file rejected

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- Replaces previous image (old file deleted on server)
- Image upload does NOT flip `is_profile_completed` by itself (but does raise `profile_completion_percentage` if image was missing)

```
const form = new FormData();
form.append("file", selectedFile);
await fetch(`${API_BASE}/doctor/profile/image`, {
  method: "POST",
  headers: { Authorization: `Bearer ${token}` },
  // Do NOT set Content-Type manually - browser sets multipart boundary
  body: form
});

// Success response
{
  "message": "Profile image uploaded successfully",
  "profile_image_url": "/uploads/doctor_image/<uuid>.jpg"
}
```

Display the image as: `{API_BASE}{profile_image_url}` e.g. `http://localhost:8000/uploads/doctor_image/....jpg` (static files are mounted at `/uploads`).

- 400 - Missing filename, wrong type, empty, > 5 MB, invalid path
- 404 - Profile not found
- 500 - Failed to save file

2.5 DELETE /doctor/profile/image

Request body: none.

```
{
  "message": "Profile image deleted successfully",
  "profile_image_url": null
}
```

- 404 - "No profile image to delete" or profile missing

2.6 Suggested Profile UI screens

- Profile Overview: avatar, name, department.name, specialization, fee, shift, completion badge + progress bar
- Edit Professional: qualification, experience_years, bio, languages (tag input)
- Edit Contact: phone + phone_code, address fields, DOB, gender select, emergency name + phone
- Avatar: crop/preview optional; enforce type + 5 MB client-side before upload
- On load: GET profile once; show spinner; if 404 show Contact admin
- On save: PUT only dirty fields (nested partial OK); refresh form from response
- Show incomplete banner when `is_profile_completed === false`
- Bind progress bar to `profile_completion_percentage`

2.7 TypeScript types (Profile)

```
export type GenderCode = 1 | 2 | 3 | 4;

export interface AddressInfo {
  line: string | null;
  city: string | null;
  state: string | null;
}
```

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```
export interface EmergencyContactInfo {
  name: string | null;
  phone: string | null;
}

export interface DepartmentInfo { id: number; name: string; }
export interface RoleInfo { id: number; name: string; }
export interface ShiftInfo {
  name: string | null;
  start_time: string | null; // "HH:MM"
  end_time: string | null;
}

export interface DoctorProfile {
  user_id: number;
  first_name: string;
  last_name: string | null;
  email: string;
  phone: string | null;
  phone_code: string | null;
  address: AddressInfo;
  date_of_birth: string | null; // YYYY-MM-DD
  gender: GenderCode | null;
  emergency_contact: EmergencyContactInfo;
  department: DepartmentInfo | null;
  role: RoleInfo | null;
  specialization: string | null;
  qualification: string | null;
  medical_license_number: string | null;
  employee_id: string | null;
  experience_years: number | null;
  joining_date: string | null;
  consultation_fee: number | null;
  bio: string | null;
  languages: string[];
  shift: ShiftInfo | null;
  profile_image_url: string | null;
  is_profile_completed: boolean;
  profile_completion_percentage: number;
  is_active: boolean;
  last_login: string | null;
  created_at: string | null;
  updated_at: string | null;
}

export interface DoctorProfileUpdate {
  qualification?: string | null;
  experience_years?: number | null; // 0-60
  bio?: string | null;
  languages?: string[] | null;
  phone?: string | null;
  phone_code?: string | null;
  address?: Partial<AddressInfo> | null;
  date_of_birth?: string | null;
  gender?: GenderCode | null;
  emergency_contact?: Partial<EmergencyContactInfo> | null;
}
```

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```
export interface DoctorProfileImageResponse {  
  message: string;  
  profile_image_url: string | null;  
}
```

3. Module B - Doctor Notifications

Prefix: /doctor/notifications | Router: Routers/doctor_notification_router.py | Tag: Doctor Notifications

In-app notifications for the logged-in doctor only (scoped by user_id). There is no WebSocket/SSE/push - poll unread-count for the nav badge. Doctors cannot create notifications; they are produced by OPD, Lab, Nurse, Admin flows.

3.1 Endpoint summary

Method + Path	Status	Permission	Purpose
GET /doctor/notifications	200	notifications:view	Paginated list + filters
GET .../unread-count	200	notifications:view	Badge count
PATCH .../{id}/read	200	notifications:update	Mark one as read
PATCH .../read-all	200	notifications:update	Mark all as read

Not implemented (do not build UI for these yet)

- Delete notification
- Preferences / mute / channels
- Push (FCM/APNs) or real-time WebSocket/SSE
- Doctor-facing create API

3.2 GET /doctor/notifications

Query parameters:

Param	Type	Rules
page	int	>= 1, default 1
limit	int	1-100, default 20
search	string	min length 1; ILIKE on title, message, created_by_name
is_read	bool	Optional filter true/false
source_module	enum	See SourceModule
notification_type	enum	See NotificationType
start_date	date	YYYY-MM-DD, inclusive (Asia/Kolkata)
end_date	date	YYYY-MM-DD, inclusive (Asia/Kolkata)

Sort order: priority CRITICAL > HIGH > NORMAL, then created_at descending.

```
GET /doctor/notifications?page=1&limit=20&is_read=false
```

```
{
  "total": 42,
  "page": 1,
  "limit": 20,
  "items": [
    {
      "id": 101,
      "user_id": 12,
      "title": "Lab Report Ready",
      "message": "CBC - Ramesh Patil",
      "notification_type": "LAB_REPORT_READY",
      "priority": "HIGH",
      "source_module": "LAB",
      "reference_type": "LAB_ORDER",
```


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```
{
  "reference_id": 55,
  "created_by": 8,
  "created_by_name": "Lab Tech One",
  "is_read": false,
  "read_at": null,
  "created_at": "2026-07-13T09:15:00+05:30"
}
```

NotificationResponse fields:

Field	Type	Notes
id	number	Primary key
user_id	number	Recipient doctor user id
title	string	Max 255
message	string null	May contain \n - render multiline
notification_type	enum string	See types table
priority	enum string	NORMAL HIGH CRITICAL
source_module	enum string	Producer module
reference_type	enum string	Deep-link entity type
reference_id	number	Deep-link entity id
created_by	number null	Actor user id
created_by_name	string null	Display name snapshot
is_read	boolean	
read_at	datetime null	ISO-8601 with timezone
created_at	datetime	ISO-8601 with timezone

3.3 GET /doctor/notifications/unread-count

```
{ "count": 5 }
```

Use for the nav bell badge. Poll every 15-30 seconds (or on focus/visibility change). No real-time channel exists.

3.4 PATCH /doctor/notifications/{notification_id}/read

Body: none. Returns full NotificationResponse. Idempotent if already read.

- Sets is_read=true and read_at=now (Asia/Kolkata) on first mark
- 404 - "Notification not found" (wrong id or not owned by current user)

3.5 PATCH /doctor/notifications/read-all

Body: none.

```
{ "message": "All notifications marked as read" }
```

After success, set local unread count to 0 and refresh the list if open.

3.6 Enums

NotificationPriority: NORMAL | HIGH | CRITICAL

Default priority by type:

- NEW_APPOINTMENT -> NORMAL
- LAB_REPORT_READY, APPOINTMENT_CANCELLED, APPOINTMENT_RESCHEDULED, ADMIN_UPDATE, HANDOVER_TAKEN_OVER, SHIFT_UPDATED -> HIGH

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- EMERGENCY_ALERT -> CRITICAL

NotificationType (values that currently appear for doctors):

Type	When created
NEW_APPOINTMENT	Paid patient added to doctor queue (check-in)
APPOINTMENT_CANCELLED	Appointment cancelled AND visit paid
APPOINTMENT_RESCHEDULED	scheduled_at changed, not cancelled, paid
LAB_REPORT_READY	Lab report uploaded / first file report created
ADMIN_UPDATE	Admin dept change / deactivate / delete / admin edits

Reserved / other enum values (exist; may appear in filters or future flows):

- PATIENT_CHECKED_IN, LAB_REPORT_UPDATED, PRESCRIPTION_CREATED, PRESCRIPTION_UPDATED
- EMERGENCY_ALERT (mainly nurse flows; doctors may still receive in some cases)
- HANDOVER_TAKEN_OVER, SHIFT_UPDATED (primarily nurse; include in TS union)

SourceModule: OPD_BILLING | LAB | NURSE | ADMIN (active for doctors). Also defined: RECEPTIONIST, PHARMACY, SYSTEM

ReferenceType (deep-link):

- APPOINTMENT + reference_id -> appointment / queue / calendar screen
- LAB_ORDER + reference_id -> lab order detail
- PATIENT + reference_id -> patient chart / emergency context
- USER + reference_id -> profile / account notice (admin updates)
- Also defined: PRESCRIPTION, BILL, SCHEDULE, LEAVE, HANDOVER, ALERT

3.7 Notification trigger details (for UI copy)

- NEW_APPOINTMENT title: "Paid Appointment Confirmed" - patient, time, token
- APPOINTMENT_CANCELLED title: "Appointment Cancelled" - patient + time
- APPOINTMENT_RESCHEDULED title: "Appointment Rescheduled" - patient + new time
- LAB_REPORT_READY title: "Lab Report Ready" - "{test}" - {patient}"
- ADMIN_UPDATE - e.g. Department reassigned, Account disabled by admin
- Unpaid appointments do NOT notify on cancel/reschedule
- Replacing an existing lab file report does NOT create another notification
- Message often contains newlines (\n) - render with white-space: pre-line

3.8 Suggested Notifications UI

- Nav: bell icon + unread badge from unread-count (poll)
- Inbox list: title, message (multiline), priority chip, type, relative time, unread style
- Filters: Unread / All, type chips, date range, search box
- Actions: click row -> mark read + navigate via reference_type/reference_id
- Toolbar: "Mark all as read"
- Empty state when total === 0
- Visual priority: CRITICAL red, HIGH amber, NORMAL default
- Do not invent delete/archive/preferences UI until backend supports it

3.9 TypeScript types (Notifications)

```
export type NotificationPriority = "NORMAL" | "HIGH" | "CRITICAL";

export type NotificationType =
  | "NEW_APPOINTMENT"
  | "APPOINTMENT_CANCELLED"
```

```

| "APPOINTMENT_RESCHEDULED"
| "PATIENT_CHECKED_IN"
| "LAB_REPORT_READY"
| "LAB_REPORT_UPDATED"
| "PRESCRIPTION_CREATED"
| "PRESCRIPTION_UPDATED"
| "EMERGENCY_ALERT"
| "ADMIN_UPDATE"
| "HANDOVER_TAKEN_OVER"
| "SHIFT_UPDATED";

export type SourceModule =
  | "OPD_BILLING" | "LAB" | "RECEPTIONIST" | "NURSE"
  | "PHARMACY" | "ADMIN" | "SYSTEM";

export type ReferenceType =
  | "APPOINTMENT" | "LAB_ORDER" | "PRESCRIPTION" | "BILL"
  | "PATIENT" | "USER" | "SCHEDULE" | "LEAVE"
  | "HANDOVER" | "ALERT";

export interface Notification {
  id: number;
  user_id: number;
  title: string;
  message: string | null;
  notification_type: NotificationType;
  priority: NotificationPriority;
  source_module: SourceModule;
  reference_type: ReferenceType;
  reference_id: number;
  created_by: number | null;
  created_by_name: string | null;
  is_read: boolean;
  read_at: string | null;
  created_at: string;
}

export interface NotificationListResponse {
  total: number;
  page: number;
  limit: number;
  items: Notification[];
}

export interface UnreadCountResponse {
  count: number;
}

```

4. Error Handling Cheat Sheet

HTTP	When / Frontend action
400	Empty profile PUT; bad image upload - show toast from detail
401	Redirect to login; clear token
403	Permission denied / inactive / not doctor - show access denied
404	Profile missing (contact admin); notification not found; no image
422	Validation failed / forbidden extra fields - show field errors
500	Image save failure - retry later

FastAPI usually returns { "detail": "..."} or a validation array for 422. Surface detail in toasts; map 422 to form fields when possible.

5. Frontend Implementation Checklist

Profile

- [] Route/page: Doctor Profile under authenticated doctor layout
- [] GET /doctor/profile on mount
- [] Separate editable vs read-only sections visually
- [] Bind nested address + emergency_contact correctly (not flat strings)
- [] Gender select mapped to 1-4
- [] Languages as chip/tag input
- [] PUT with only changed fields; nested partial objects OK
- [] Avatar upload via FormData field "file"; preview + 5 MB client check
- [] Avatar delete + clear preview
- [] Show is_profile_completed banner / CTA until true
- [] Progress bar from profile_completion_percentage
- [] Image URL = API_BASE + profile_image_url
- [] Display shift, employee_id, joining_date as read-only if present

Notifications

- [] Bell in doctor header with unread badge
- [] Poll GET /doctor/notifications/unread-count (15-30s)
- [] Notifications page/drawer with pagination
- [] Filters: is_read, notification_type, date range, search
- [] Mark one read on open; Mark all read button
- [] Deep-link by reference_type + reference_id
- [] Render message with line breaks; priority colours
- [] Optimistic unread decrement after mark-read

6. Permissions Required (Doctor role)

- doctor_profile:view
- doctor_profile:update
- doctor_profile:upload_image
- doctor_profile:delete_image
- notifications:view

- notifications:update

These are seeded for the doctor role. Permissions are embedded in the JWT at login. Gate UI routes/buttons on permissions if your app checks them client-side.

7. Quick API Reference

```
# Profile
GET    /doctor/profile
PUT    /doctor/profile
POST   /doctor/profile/image      (multipart: file)
DELETE /doctor/profile/image

# Notifications
GET    /doctor/notifications
GET    /doctor/notifications/unread-count
PATCH /doctor/notifications/{id}/read
PATCH /doctor/notifications/read-all

# Auth
Authorization: Bearer <token>
OpenAPI: /docs when backend is running
```

8. Source Files (Backend)

- Routers/doctor_profile_router.py
- Schemas/doctor_profile_schema.py
- Services/doctor_profile_service.py
- Models/doctor_profile.py
- Routers/doctor_notification_router.py
- Schemas/notification_schema.py
- Enums/notification.py
- Models/notification.py
- Services/notification_service.py
- seed.py (permissions)

End of document (v2.0). Same content as doctor-profile-notifications-frontend-guide.docx. Regenerate: `python Docs/generate_doctor_profile_notification_pdf.py`